

modern linux cli tools

Patrick Drechsler

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- software developer
- ☀️ dotnet / C#
- 🌙 Linux on the desktop 😎
- slides are online: see QR-Code





"modern" is relative...

Douglas Adams

I've come up with a set of rules that describe our reactions to technologies:

- Anything that is in the world **when you're born** is normal and ordinary and is just a natural part of the way the world works.
- Anything that's invented **between when you're fifteen and thirty-five** is new and exciting and revolutionary and you can probably get a career in it.
- Anything invented **after you're thirty-five** is against the natural order of things.





Target audience

- linux desktop CLI users
- linux admins
- docker users
- devops/cloud users
- ...

```
-----  
/ This is the year of linux on the desktop \  
|  
|      ...Windows10 has WSL ;-)  
|  
\  
-----  
  
  \  ^__^  
  \  (oo)\_____  
    (__)\\       )\/\  
        ||----w |  
        ||     ||
```



Linux tooling philosophy

- do one thing (and only one thing!) well
- chaining
- *sound familiar? (hint: functional programming...)*



Why? Improvements to...

■ productivity

- one cli command is better than multiple clicks in a GUI
- cli commands are reproducible and easier to document (no screenshots: remember IIS?)

■ optics

- "unix porn" (``PS1`` , ``ls`` , ...)

Learn the basics

VS Code is cool, but it is not a CLI tool

■ Emacs and Vim:

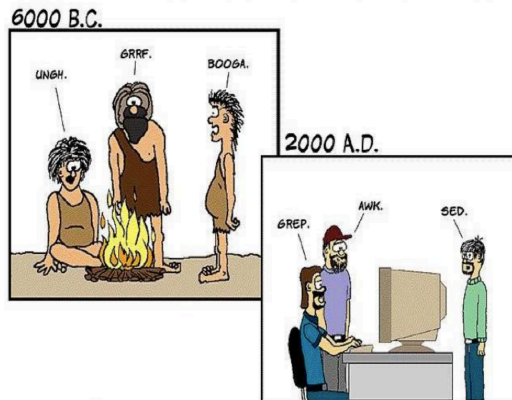
- navigation within file
- cut, copy & paste
- navigation between files
- and how to exit
- pick one and become fluent

■ Learn basic cli commands: ``cd``, ``ls``, ``cat``, ``less``, ``find``, ``grep``, ``sed``, ``tail``, ``awk``, ``dd``, ``rsync``, ...

■ Combine using pipes and redirection



EVOLUTION OF LANGUAGE THROUGH THE AGES.





Pets

Cattle





terminal and shell: what's the difference?



terminal

- **colors** (16 or more),
- **fonts** (utf8, ligatures, emojis, ...)
- **interactions**
 - keyboard shortcuts
 - mouse interaction (copy & paste, scrolling, selection, ...)

shell

- **everything else** (f.ex. ``PS1``, scripting language)

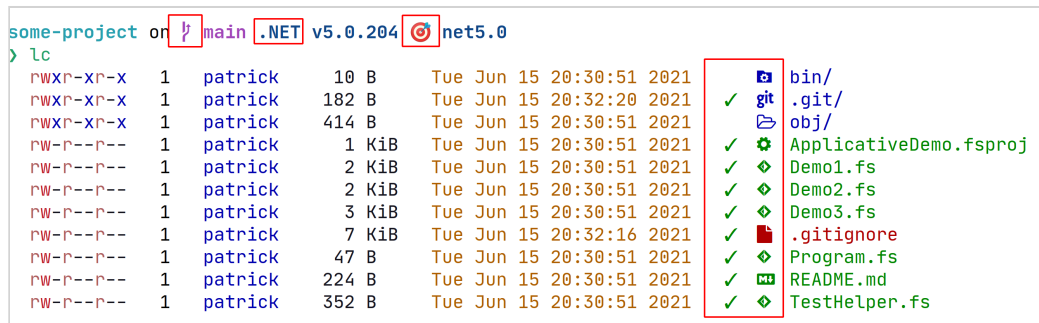
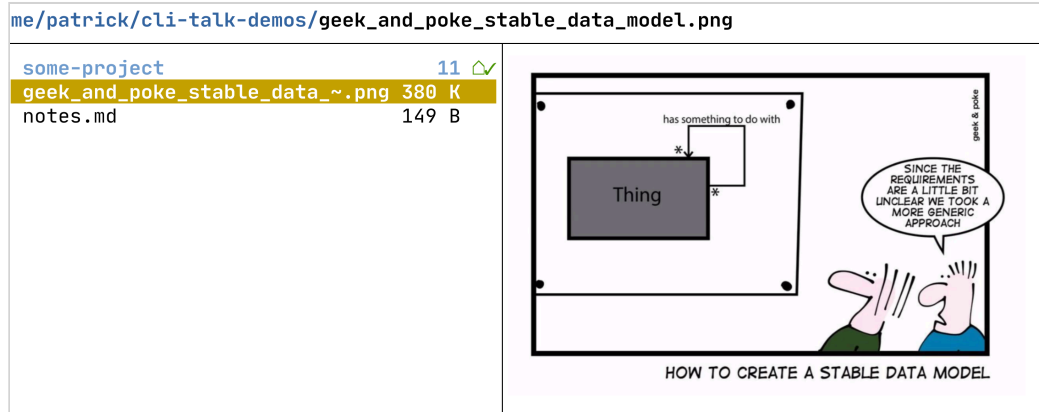


What is the best terminal?

- start by using your default terminal

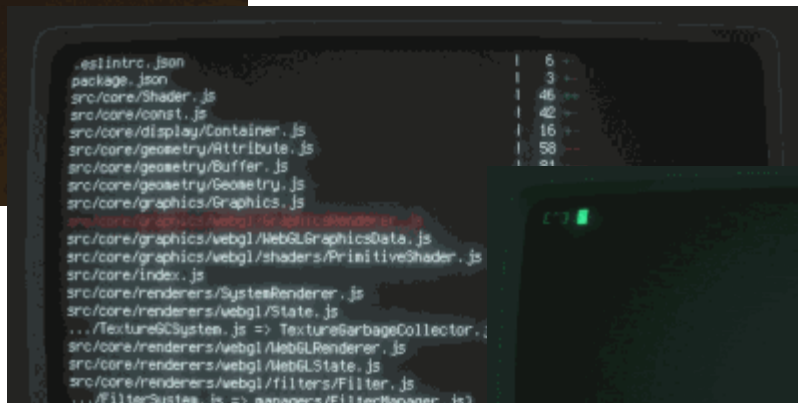
but think about enriching your output!

- Emojis: 👍 ✅ 💥 🤞 💩 € ★
- Unicode, Emojis, Fonts, Image support (!)



terminal - Retro style

■ cool-retro-term: 🌟 18k stars <https://github.com/Swordfish90/cool-retro-term>





modern alternatives

not the topic of this talk, but...

- kitty: 🌟 20k stars <https://github.com/kovidgoyal/kitty>
- alacritty 🌟 49k stars <https://github.com/alacritty/alacritty>
- wezterm 🌟 10k stars <https://github.com/wez/wezterm>



Popular shells

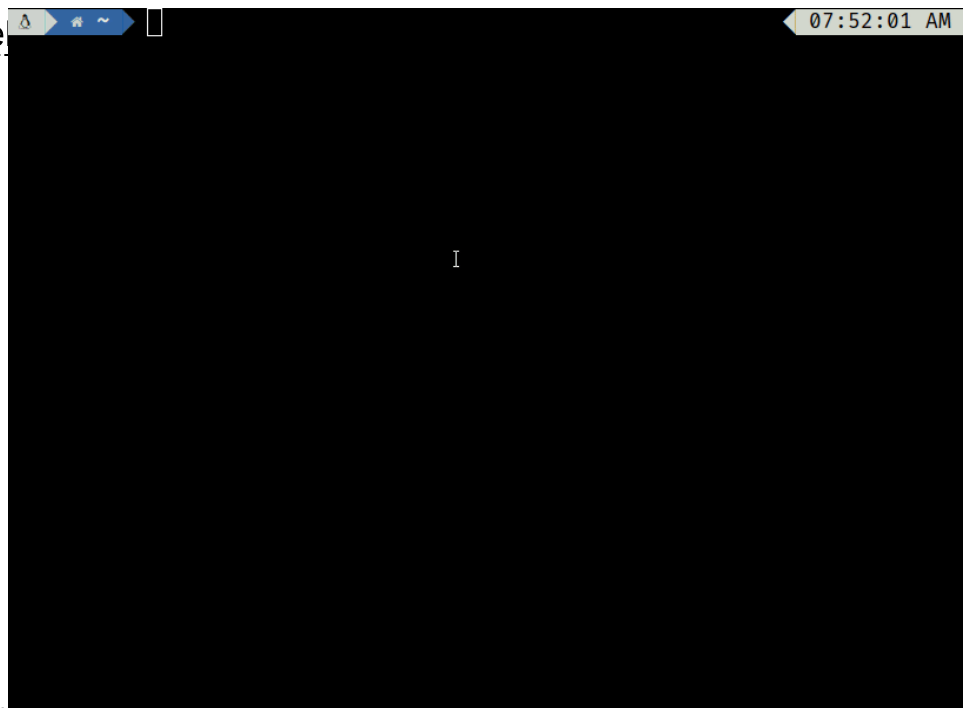
- bash
- zsh
- fish
- dash (embedded)



nushell: The new kid on the block

Everything is data

- keep an eye on it!
 - 26k stars <https://github.com/nushell/nushell>
- 100% **NOT** POSIX conform
- category: experimental





PS1: alias for "prompt string 1"

- default: ``username@machinename:~/some/location $``
- ``$PWD`` present working directory

Extra, nice-to-have information

- only show relevant parts (hide user/machine name on local machine)
- compact location (``$PWD``)
- git status
- environment infos (node / .NET / Java / Python / etc version)

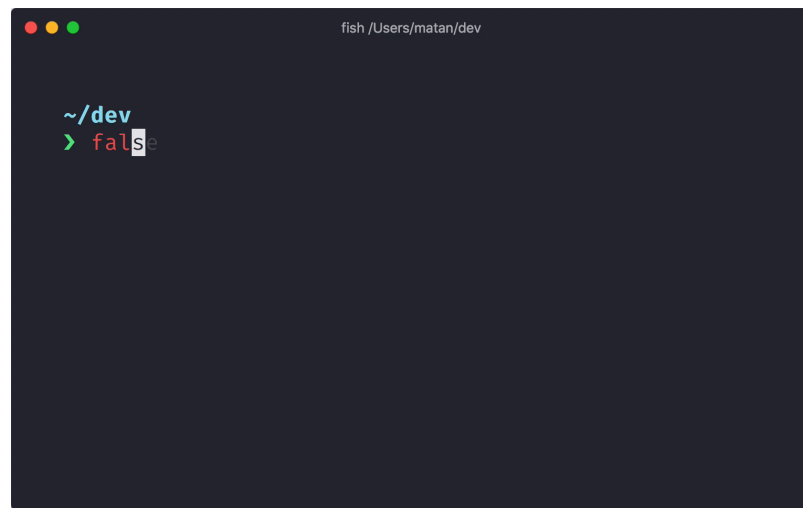
PS1: Starship

the new kid on the block

- 33k stars <https://github.com/starship/starship>
- cross-platform, cross-shell (even Windows cmd.exe)
- stylish
- great defaults for many environments
- easy to configure
- can look like powerline
- category: unix-porn

starship
CROSS-SHELL PROMPT

The minimal, blazing-fast, and infinitely customizable prompt for any shell!





terminal-independent shell UI

- 28k stars <https://github.com/tmux/tmux>
- terminal multiplexer
- "tabs" and "split screen" features: terminal independent
- detachable sessions
- plugin system & package manager (of course)
- customizable
- scriptable: f. ex. `tmuxinator``



``tmux`` -wrapper for non-vim users

- (mirrored repo: ~1k stars) <https://www.byobu.org/>
- Keybindings ``F1`` - ``F12``
- Sensible defaults (layout, info line)
- category: productivity



Warning about tiling everything



don't forget about all the fun keybindings

Nesting can become a challenge...

Abstraction level	Keybinding Brain Overload
tiling window manager like <code>`i3`</code> or <code>`sway`</code>	🧠
tiling terminal like <code>`kitty`</code>	🧠🧠
tiling shell like <code>`nushell`</code> or <code>`tmux`</code>	🧠🧠🧠
application like <code>`emacs`</code> or <code>`vim`</code>	🧠🧠🧠🧠



Smooth git handover

- 1k stars <https://mob.sh/>
- remote ensemble or pair programming using screen sharing



```
## first user
mob start 5
mob next

## second user
mob start 5
mob next

## any user, wanting to make a normal commit
mob done
```



file explorer / manager

- 13k stars <https://ranger.github.io>
- 2 layout options
 - miller columns ("mac" style)
 - multipane (similar to Midnight commander)
- powerful preview mode and program launcher (word, music, ...)
 - with image support for certain terminals (``iterms2`` , ``urxvt`` , ``kitty``) not ``gnome-terminal`` (!)
- key bindings: see `~/ .config/ranger/rc.conf`` starting at line ~300...
- category: navigation, file system



another terminal file manager

Most terminal file managers out there (if not all) are built using the TUI design principles, much like the superb ``Midnight Commander`` and ``Ranger``. But `clifm` is built rather having the CLI design principles in mind: do not navigate through a big menu of files, just type it!, exactly as you do in your regular shell, but easier and faster.

For this reason, `clifm` does not need to be better: it's just different!

- 1.1k stars <https://github.com/leo-arch/clifm>
- ``ranger`` alternative
- category: navigation, file system

15:11 ~

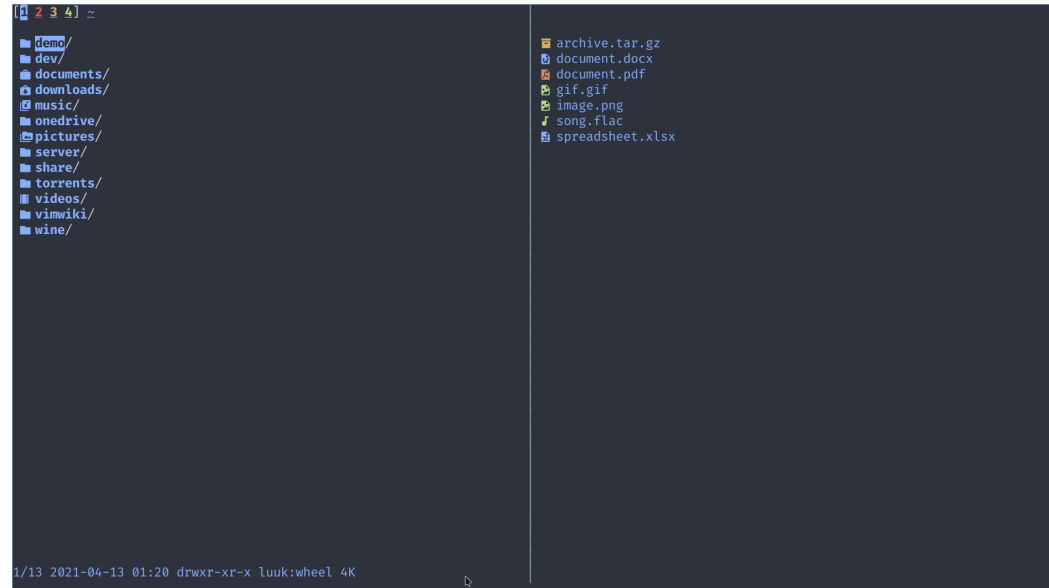
> █



another terminal file manager

nnn (n³) is a full-featured terminal file manager. It's tiny, nearly 0-config and incredibly fast.

- ``ranger`` alternative
- category: navigation, file system

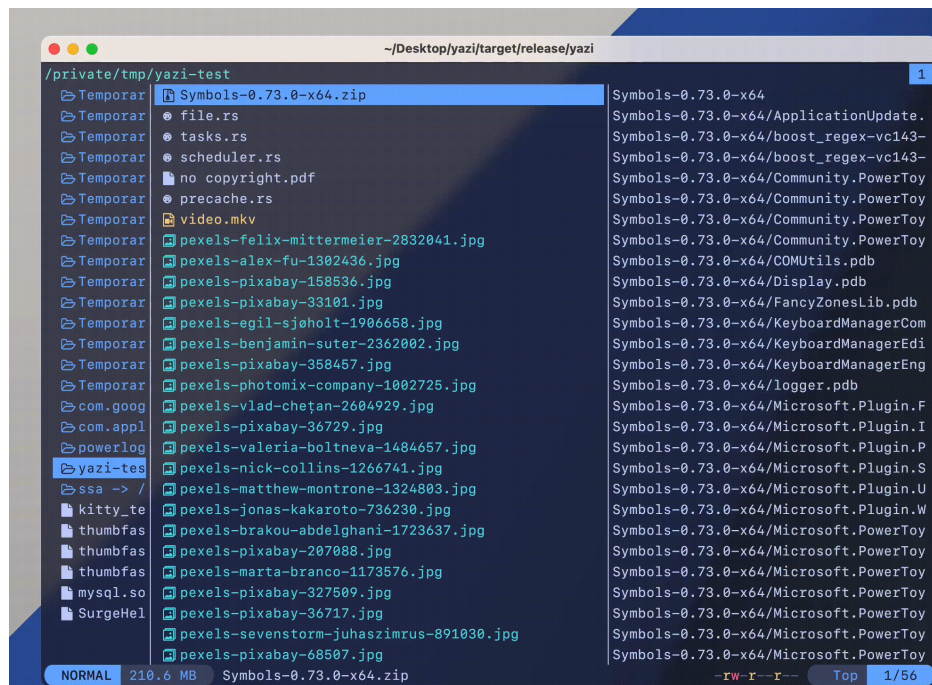




fast terminal file manager

Yazi ("duck" in Chinese) is a terminal file manager written in Rust, based on non-blocking async I/O. It aims to provide an efficient, user-friendly, and customizable file management experience.

- 3k stars <https://github.com/sxyazi/yazi>
- `ranger` alternative
- category: navigation, file system





man pages can be difficult

- 44k stars <https://tldr.sh/>
- clients implemented in many languages
 - js, ruby, python, perl, haskell, etc.
- shared user-contributed knowledge-base



- 2k stars <https://github.com/dbrgn/tealdeer>
- ...in rust

```
~/cli-talk-demos
```

```
> tldr scp
```

scp

Secure copy.

Copy files between hosts using Secure Copy Protocol over SSH.

More information: <https://man.openbsd.org/scp>.

- Copy a local file to a remote host:

```
scp path/to/local_file remote_host:path/to/remote_file
```

- Use a specific port when connecting to the remote host:

```
scp -P port path/to/local_file remote_host:path/to/remote_file
```

- Copy a file from a remote host to a local directory:

```
scp remote_host:path/to/remote_file path/to/local_directory
```

- Recursively copy the contents of a directory from a remote host to a local directory:

```
scp -r remote_host:path/to/remote_directory path/to/local_directory
```

- Copy a file between two remote hosts transferring through the local host:

```
scp -3 host1:path/to/remote_file host2:path/to/remote_directory
```

- Use a specific username when connecting to the remote host:

```
scp path/to/local_file remote_username@remote_host:path/to/remote_directory
```

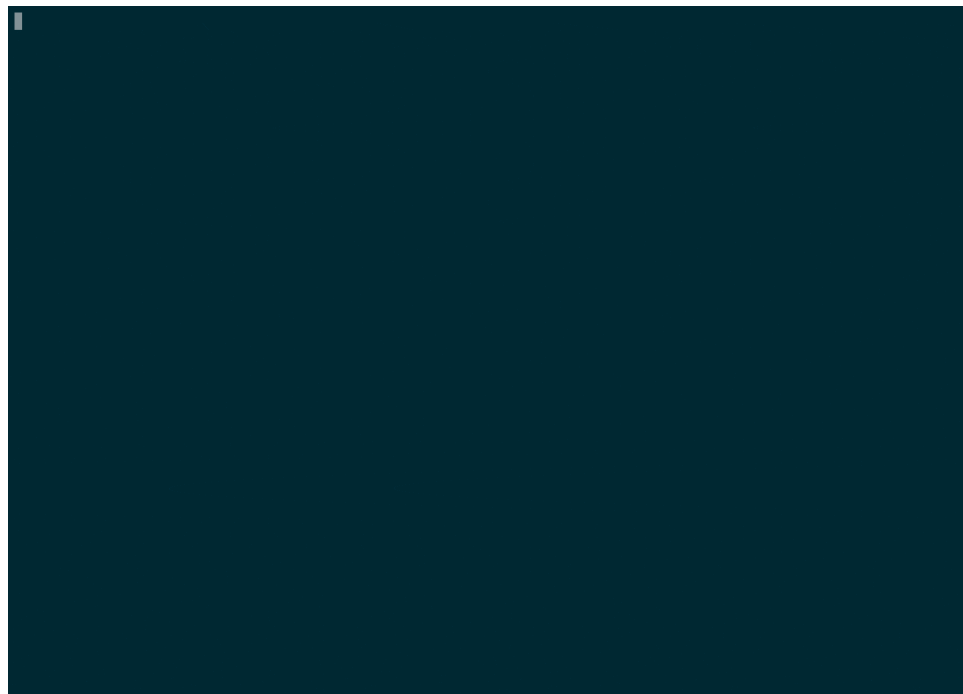
- Use a specific ssh private key for authentication with the remote host:

```
scp -i ~/.ssh/private_key local_file remote_host:path/to/remote_file
```



magic shell history

- 23k stars <https://github.com/atuinsh/atuin>
- powerful history search (`ctrl+r`)
- uses SQLite under the hood
- also allows synchronizing between machines
- cross-shell
- Example: `dotnet tool`





a smarter cd command, inspired by z and autojump.

- 26k stars <https://github.com/ajeetsouza/zoxide>
- cross-shell
- interactive mode
- integrates nicely with other tools such as ``fzf``, ``ranger``, ``yazi``, ``atuin``



``cat`` & ``less`` with syntax highlighting

- 40k stars <https://github.com/sharkdp/bat>
- hint: can be used as preview by ``ranger`` ...
- category: file preview



very fast ``grep`` replacement

- 37k stars <https://github.com/BurntSushi/ripgrep>
- ``ripgrep`` **recursively** searches directories for a regex pattern
- sensible defaults: respect ``.gitignore``, ignores hidden files & folders
- command: ``rg``
- category: search
- use ``rg --hidden -g '!.git' "your search"`` to search all hidden folder except ``.git``
 - create an alias (maybe ``rgh`` ?)...



ripgrep, but also search in PDFs, E-Books, Office documents, zip, tar.gz, sqlite (!), etc

- 5k stars <https://github.com/phiresky/ripgrep-all>
- command: ``rga``
- faster than ``pdfgrep`` (see website for a benchmark)
- category: search
- Demo: search in Manning books folder



simple alternative to `find`

- 27k stars <https://github.com/sharkdp/fd>
- "The command name is 50% shorter than `find`"
- Convenient syntax `fd PATTERN` (instead of `find -iname '*PATTERN*'`)
- Sensible defaults: `.gitignore`, ignore hidden files/folders
- fast
- category: search



interactive fuzzy search

- 51k stars <https://github.com/junegunn/fzf>
- excellent integration with other tools: Install it and it will be used by many other tools
- ``Alt+c`` : quick ``cd`` navigation
- nice helper methods. ``<cmd> **<TAB>`` , for example:
 - ``cd **<TAB>``
 - ``ssh **<TAB>``
 - ``unset **<TAB>`` , ``export **<TAB>`` , ``unalias **<TAB>``
 - ``kill -9 <TAB>`` careful...
- Things you can do with fzf: <https://andrew-quinn.me/fzf>
- category: search



"sql for json" (and other data formats)

jq

- lightweight and flexible command-line JSON processor
- 🌟 26k stars <https://github.com/jqlang/jq>

jqp

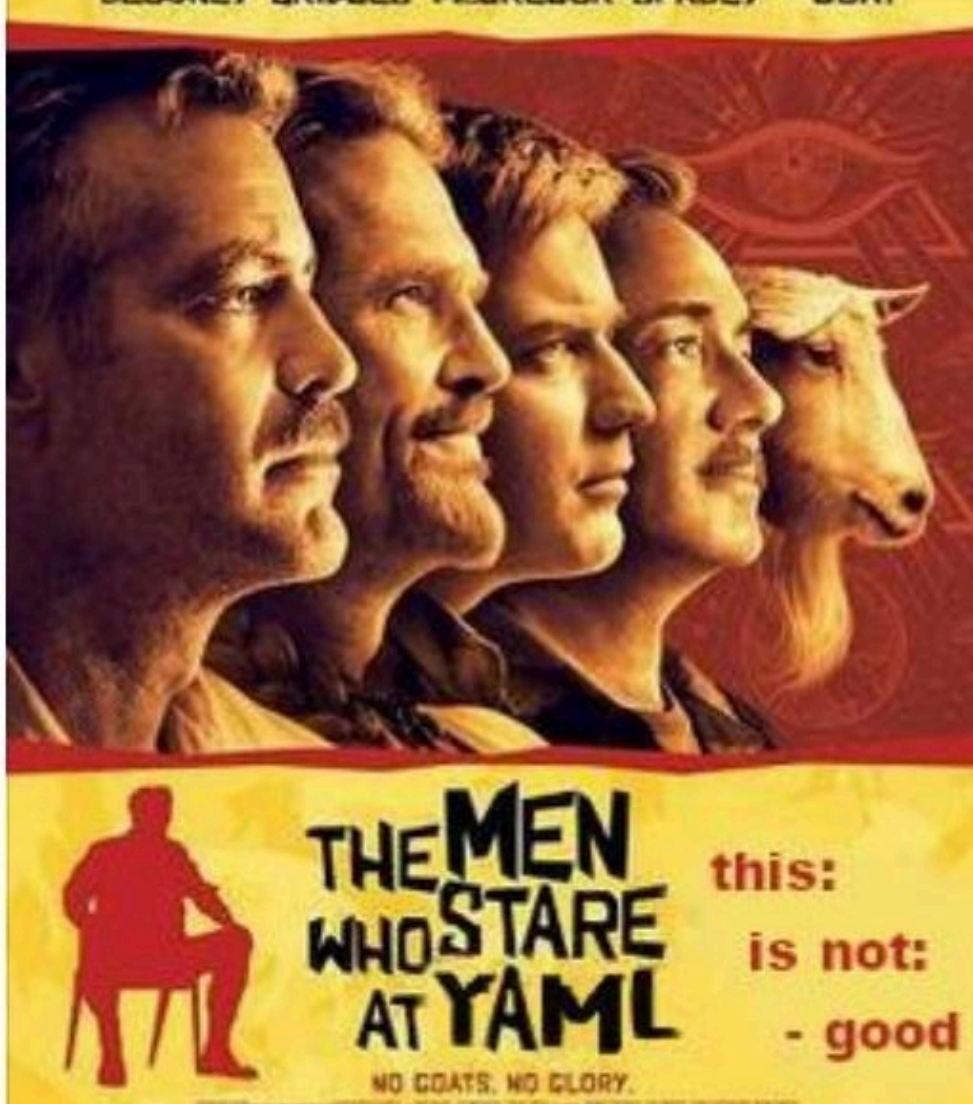
- TUI for `jq``
- 🌟 1.8k stars <https://github.com/noahgorstein/jqp>

millier

- similar to `jq``, but also for CSV, TSV, JSON, etc.
- 🌟 8k stars <https://github.com/johnkerl/miller>

fq

- `jq``, but for binary data
- 🌟 8k stars <https://github.com/wader/fq>
- for example: audio, video, images, etc.



yq

■ ``jq``, but for YAML

■ 13k stars

<https://github.com/mikefarah/yq>



benchmarking tool

■ 15k stars

<https://github.com/sharkdp/hyperfine>

```
## comparing `fd` with `find`  
hyperfine --warmup 3 \  
  'fd -e jpg -uu' \  
  'find -iname "*.jpg"'
```

■ category: benchmarking



monitor any kind of "copy"

- 6k stars <https://github.com/Xfennec/progress>
- attach to any kind of copy, even after the fact (!)
- category: monitoring

```
No command currently running: cp, mv, dd, tar, cat, rsync, grep, fgrep, egrep, cut, sort, md5sum,  
sha1sum, sha224sum, sha256sum, sha384sum, sha512sum, adb, gzip, gunzip, bzip2, bunzip2, xz,  
unxz, lzma, unlzma, 7z, 7za, zcat, bzip2, lzcat, split, pgp, or wrong permissions.
```



Rainbows and unicorns

- 5k stars <https://github.com/busyloop/lolcat>
- category: fun, unix porn

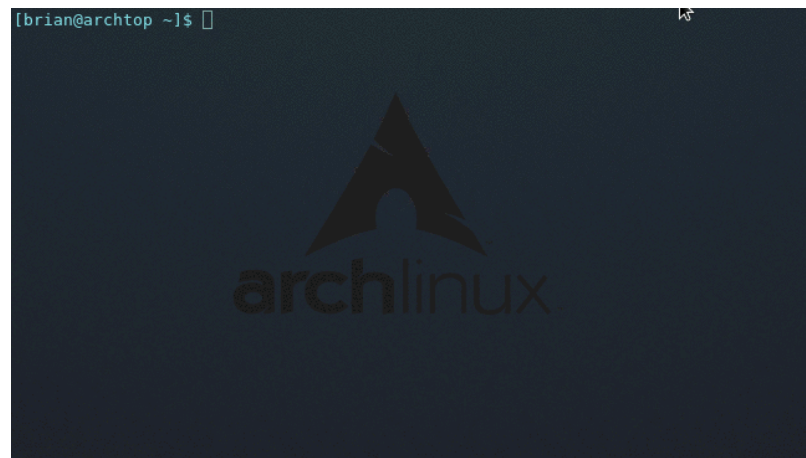
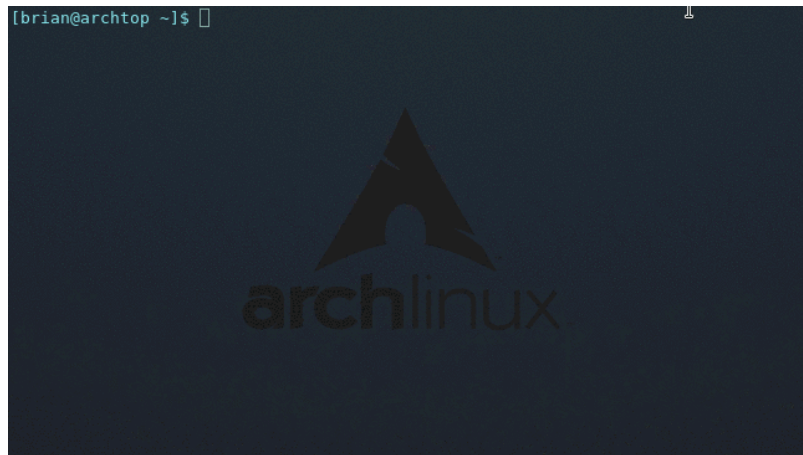
```
~/cli-talk-demos
> lc | lolcat
rwxr-xr-x 1 patrick 182 B Thu Jun 17 07:24:41 2021 search-project/
rwxr-xr-x 1 patrick 872 B Wed Jun 16 01:30:02 2021 search-project-pdf/
rwxr-xr-x 1 patrick 196 B Tue Jun 15 20:32:18 2021 some-project/
rw-r--r-- 1 patrick 379 KiB Tue Jun 15 20:39:21 2021 geek_and_poke_stable_data_model.png
rw-r--r-- 1 patrick 23 B Tue Jun 15 22:01:28 2021 live-notes.md
rw-r--r-- 1 patrick 149 B Tue Jun 15 16:40:27 2021 notes.md
rw-r--r-- 1 patrick 2 GiB Wed Jun 16 01:28:31 2021 ubuntu-20.04.2.0-desktop-amd64.iso

~/cli-talk-demos
> lc | lolcat
rwxr-xr-x 1 patrick 182 B Thu Jun 17 07:24:41 2021 search-project/
rwxr-xr-x 1 patrick 872 B Wed Jun 16 01:30:02 2021 search-project-pdf/
rwxr-xr-x 1 patrick 196 B Tue Jun 15 20:32:18 2021 some-project/
rw-r--r-- 1 patrick 379 KiB Tue Jun 15 20:39:21 2021 geek_and_poke_stable_data_model.png
rw-r--r-- 1 patrick 23 B Tue Jun 15 22:01:28 2021 live-notes.md
rw-r--r-- 1 patrick 149 B Tue Jun 15 16:40:27 2021 notes.md
rw-r--r-- 1 patrick 2 GiB Wed Jun 16 01:28:31 2021 ubuntu-20.04.2.0-desktop-amd64.iso
```



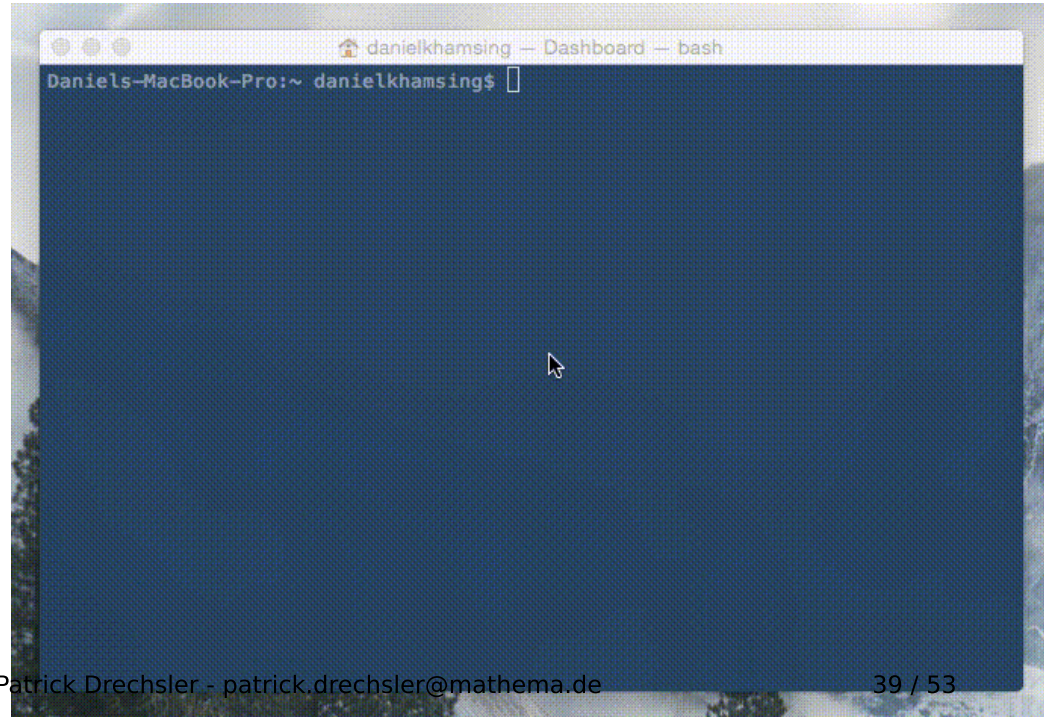
when the tv team comes in your office

- 5k stars <https://github.com/bartobri/no-more-secrets>
- pipe something to ``nms`` or just run ``sneakers``
- category: fun, unix porn





- 3k stars <https://github.com/mtoyoda/sl>
- typo ``sl`` (instead of ``ls``) → show steam locomotive
- can we pipe it to ``lolcat``? of course
- category: fun, unix porn





GitHub CLI

- 32k stars <https://github.com/cli/cli>
- Example: push current git folder to new private github repo:
 - `gh repo create --source . --push --private``
- create PRs
- view issues



static analysis tool for shell scripts

- 37k stars <https://github.com/koalaman/shellcheck>
- plugins available for most IDEs
- written in Haskell ;-)



- 12k stars <https://github.com/go-task/task>
- task runner / simpler Make alternative
- single binary
- cross-platform
- written in Go

```
version: '3'

vars:
  GREETING: Hello, World!

tasks:
  default:
    cmds:
      - echo "${GREETING}"
    silent: true

  test:
    cmds:
      - dotnet test
```



Command suggestions, command-not-found and ``thefuck`` replacement written in Rust

- 200 stars <https://github.com/iffse/pay-respects>
- category: productivity



ls on steroids

``ls`` problem: **sort by name and time and size** at the same time...

Use colors!

- ``colorls`` ("the original" in ruby: <https://github.com/athityakumar/colorls>)
- ``lsd`` (in rust: <https://github.com/Peltoche/lsd>)
- ``exa`` (in rust: <https://github.com/ogham/exa>)
- ``eza`` (in rust: <https://github.com/eza-community/eza>)

Required: font providing all symbols

- Example: NerdFonts <https://github.com/ryanoasis/nerd-fonts>



- 7k stars <https://github.com/eza-community/eza>
- category: unix porn

```
drwxr-xr-x - patrick patrick 17 Jun 2023 | gh-pages | 2023-06-cloudland-modern-linux-cli-tools
drwxr-xr-x - patrick patrick 4 Jun 2023 | + main | 2023-06-dwx-fp-csharp-to-fsharp
drwxr-xr-x - patrick patrick 23 Jun 2023 | gh-pages | 2023-06-my-coding-zone-error-handling
drwxr-xr-x - patrick patrick 17 Nov 2023 | gh-pages | 2023-06-socrates-day-franken-modern-linux-cli-tools
drwxr-xr-x - patrick patrick 12 Dec 2023 | it-tage-. | 2023-12-it-tage-fp-intro-csharp-fsharp
drwxr-xr-x - patrick patrick 10 Jan 05:35 | | 2023-fp-intro-csharp-fsharp
drwxr-xr-x - patrick patrick 4 May 2023 | main | 2023-gregor
drwxr-xr-x - patrick patrick 28 Apr 22:50 | 2024-mag.. | 2023-mathema-sample-slidev
drwxr-xr-x - patrick patrick 13 Apr 02:03 | main | 2023-modern-linux-cli-tools
drwxr-xr-x - patrick patrick 6 Jul 2022 | - | 2024-04-mathemacampus-approval-testing
drwxr-xr-x - patrick patrick 19 Feb 21:18 | main | _other_people
drwxr-xr-x - patrick patrick 8 Jan 02:58 | master | abstracts
drwxr-xr-x - patrick patrick 26 Jun 2023 | main | abstracts.orig
drwxr-xr-x - patrick patrick 22 Oct 2019 | - | draptik
drwxr-xr-x - patrick patrick 3 Apr 2023 | main | ideas-workshop-testing
drwxr-xr-x - patrick patrick 8 Feb 2023 | main | jeopardy-questions
drwxr-xr-x - patrick patrick 10 Dec 2018 | master | mathema-teamday-openspace
drwxr-xr-x - patrick patrick 31 Aug 2020 | - | reveal-js-template
drwxr-xr-x - patrick patrick 5 Nov 2023 | - | template
drwxr-xr-x - patrick patrick 27 Aug 2018 | - | tmp
-rw-r--r- 87k patrick patrick 14 Feb 09:33 | - | drechsler-profile.jpg
-rw-r--r- 9.5M patrick patrick 14 Feb 09:33 | - | 'Mathema Campus-ppt-Vorlage.pptx'
```



Everything lazy*

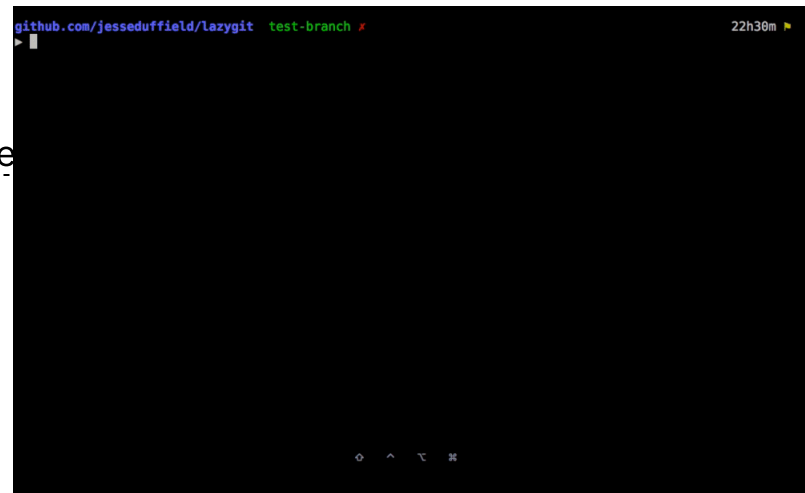
■ lazydocker

■ 27k stars <https://github.com/jesseduffield/lazydocker>

■ lazygit

■ 35k stars <https://github.com/jesseduffield/lazygit>

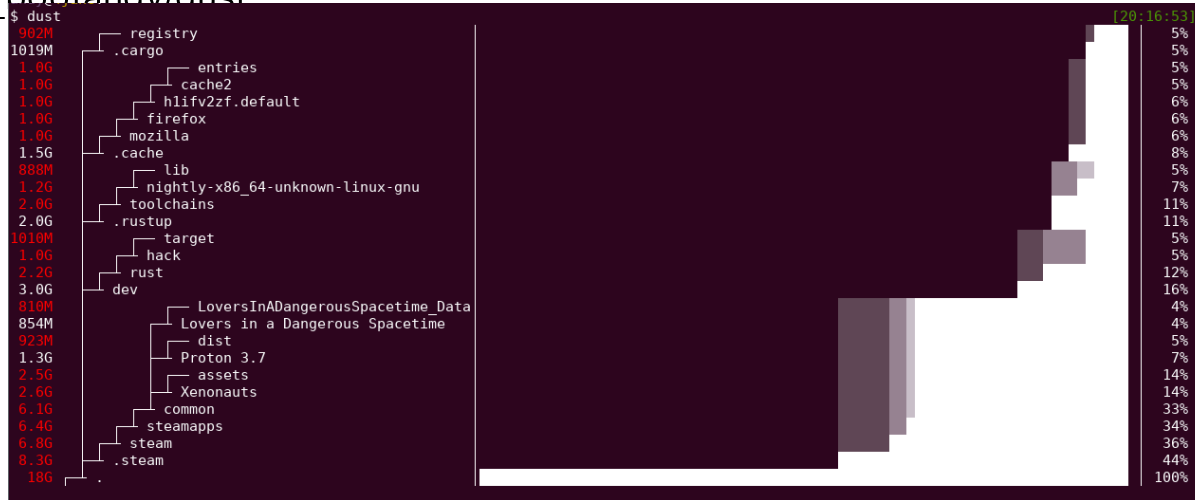
-Status-		-Config-	
lazydocker		Name: <none>	
-Services-		ID: sha256:a28c7b34cac8c23430c15a6c1a6738e998859057848457cf05	
running	api 0.01%	Tags: <none>:<none>	
running	mysql 0.13%	Size: 144.02MB	
running	quote-fetcher 0.00%	Created: Sat, 29 Jun 2019 15:30:43 AEST	
running	redis4.0 0.31%		
-Standalone Containers-			
exited (1)	migrate 0.00%		
running	guard 0.00%		
running	rails-console 0.00%		
-Images-			
<none>	<none>	1	
<none>	<none>	1	
bumper	latest	9	
<none>	<none>	5	
-Volumes-			
local	68473898e49e0e4d2c5998f68c79	ID	TAG
local	90d4357f8a93810ee8f2a59671e7	a28c7b34ca	SIZE
local	b9a4a32a2d4d6366f9a9434aff8a	fa4b5d3656	COMMAND
local	bc80a55695b8d780db6f8e54f31a	39923e2d34	679.20kiB COPY file:1dc4cce0e59430462e9277b
		62b1340936	2.51kiB COPY file:023433ee488fb0be495da22
		b4016cf6d7	232.00B COPY file:21bfa394d874f9add12dd13
		ce36200b46	0B WORKDIR /app
		914bfdbef6	0B ENV NODE_ENV=development
		node:10-slim	0B ARG FURY_TOKEN
		<missing>	0B CMD ["node"]
		<missing>	4.83MiB /bin/sh -c set -ex && for key i
		<missing>	78 ENV YARN_VERSION=1.13.0
		<missing>	78.81MiB /bin/sh -c buildDeps='xz-utils'
		<missing>	0B ENV NODE_VERSION=10.15.3
		<missing>	325.54kiB /bin/sh -c groupadd --gid 1000 no
		<missing>	0B CMD ["bash"]
		<missing>	52.72MiB ADD file:4fc310c0cb879c876c5c0f57



du alternatives

``du`` in the name

- `ncdu` - we all know this
 - <https://dev.yorhel.nl/ncdu>
- 2k stars `gdu`: <https://github.com/dundee/gdu>
- 6k stars `dust`: <https://github.com/bootandy/dust>





- <https://github.com/dustinkirkland/hollywood>
- random crazy stuff in the terminal
- category: unix porn
- Live demo



Everything *top

- btop 🕶️
- apachetop
- ngxtop
- mtp
- pg_top
- powertop
- iotop
- iftop
- nethogs



Why didn't I show this presentation in a shell?

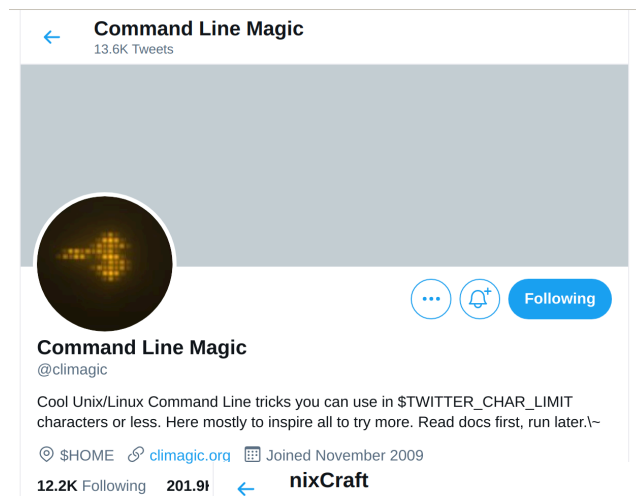
I've done it 😇

■ 🔖 5k stars <https://github.com/mfontanini/presenterm>

Resources

-  <https://mastodon.social/@climagic>
-  <https://mastodon.social/@nixCraft>
-  [r/commandline](https://www.reddit.com/r/commandline)
-  [r/unixporn](https://www.reddit.com/r/unixporn)
- search for "**awesome-cli**"

This talk was inspired by Martin Leyrer's session at [Gulaschprogrammierenacht 19 \(CCC\)](https://www.ccc-congress.org/).





I forgot your favorite tool?

- git-fuzzy <https://github.com/bigH/git-fuzzy>
- forgit <https://github.com/wfxr/forgit>
- mosh (robust replacement for ssh) <https://mosh.org/>
- fish_config (web-based configuration tool for fish shell) https://fishshell.com/docs/current/commands.html#fish_config
- xclip (cli to X clipboard) 403 stars <https://github.com/astrand/xclip>
- neofetch (fancy system info in the shell) 7.7k stars <https://github.com/dylanaraps/neofetch>
- expect <https://likegeeks.com/expect-command/>
- httpie (cli http client) 45k stars <https://httpie.org/>
- http-prompt (cli http client) 7.6k stars <http://http-prompt.com/>
- pywal (color schemes for terminal) 3.4k stars <https://github.com/dylanaraps/pywal>
- ytfzf <https://github.com/pystardust/ytfzf> find YT videos and downloads them using youtube-dl
- figlet <http://www.figlet.org/> ascii art
- toilet <http://caca.zoy.org/wiki/toilet> ascii art
- apt-iselect (interactive ncurses package search for debian) <https://www.rot13.org/~dpavlin/apt-iselect.html>
- gcalcli (google calendar in the shell) 2k stars <https://github.com/insanum/gcalcli>
- pastebinit (cli tool to send data to pastebin) 17 stars <https://github.com/skorokithakis/pastebinit>
- MapSCII (ascii google maps in the terminal) <https://github.com/rastapasta/mapscii>
- alwaysontop (always move prompt to top of screen) 160 stars <https://github.com/swirepe/alwaysontop>
- wttr.in (ascii weather report) <https://github.com/chubin/wttr.in>
- gpt4free <https://github.com/xtekky/gpt4free>
- Fig <https://fig.io/> Autocomplete on-steroids
- xkcd.xsh <https://gist.github.com/Lassi-Koykka/9fb934732a871ca3c8bc9396983a3310>
- fx (json cli viewer) <https://github.com/antonmedv/fx>



Thank You!

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Slides 

- QR Code or
- <https://draptik.github.io/2025-05-mathema-campus-modern-linux-cli-tools>

